

Daniel W Liu

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EDUCATION

University of Waterloo | *Double Major in Computer Science and Finance*
Bachelor of Computing and Financial Management (Co-op) | **GPA: 3.94/4.0**

Sept 2025 – April 2029
Waterloo, ON

EXPERIENCE

Software Developer Intern

May 2026 – Aug 2026

PlayStation Sony Interactive Entertainment

Waterloo, ON

- Engineered a **34-endpoint** Express.js admin REST API in an authorization proxy, replacing manual database edits and surfacing a React portal for admins to manage roles, permissions, and service tokens for **2,500+ developers**.
- Developed an authorization microservice that derives roles from GitHub teams through an RBAC role inheritance graph, automatically propagating permission changes to **12+ internal tools** on a 13-table PostgreSQL schema.
- Migrated publishing for 16 Kubernetes Helm charts from Jenkins to GitHub Actions, providing AWS ECR access through CloudFormation-defined IAM roles and **cutting publish time by 56%** with Docker build optimizations.

Software Engineering Intern

Jan 2026 – Apr 2026

Wedge YC S25

San Francisco, CA

- Shipped a full-stack autonomous voice agent platform using Next.js, PostgreSQL, and LiveKit that calls health insurers to check claim status, iterating weekly with **3 hospital clients**, fully deployed across **50+ clinical sites**.
- Designed a mapping engine that recursively re-dials insurers to explore undiscovered phone menu trees, merging every call into the fastest verified path to a human per number across a **4,839-number** registry of **2,841 payers**.

Software Engineering Intern

Sept 2025 – Dec 2025

HOSA Canada

Toronto, ON

- Architected a scalable full-stack resource sharing platform with Next.js, AWS S3, and PostgreSQL that publishes written test events through Redis Streams to serve **11,000+ Canadian students**, verified by **1,098 Jest tests**.
- Deployed a rate-limited MERN workshop sign-up service processing **19,943 registrations** across **347 sessions**, draining distributed Redis-backed job queues with PM2 workers to automate Twilio SendGrid email notifications.

Machine Learning Research Intern

Jul 2025 – Aug 2025

Pearl Lab Holland Bloorview Kids Rehabilitation Hospital

Toronto, ON

- Developed real-time computer vision for *Boottle Blast*, a gamified rehab platform clinically tested at **3+ hospitals**, fusing depth cameras with OpenCV and PyTorch CNN inference to reconstruct children's motor movement in 3D.
- Trained ML YOLOv8 detection models on a custom dataset built from **250+ environments**, benchmarked using NumPy and pandas to beat classical segmentation by **46.8% mean average precision at sub-10 ms** inference.

PROJECTS

rainbolt.ai | *Next.js, Pinecone, LangChain, Google Cloud, OAuth*

Devpost GitHub

- Awarded **Best UI Hack** at UofT Hack the Valley X among **124 projects** for an AI geolocation tool that teaches how to pinpoint an image's location using GeoGuessr documentation, built with Next.js, Firebase, and Zustand.
- Architected a LangChain RAG pipeline embedding street-level imagery with OpenAI CLIP into a Pinecone index of **955,199 image vectors**, retrieving matches through a FastAPI backend to ground each guess in visual evidence.

Pomme | *C++20, React.js, Express.js, Socket.IO, Arduino UNO R*

Devpost GitHub

- Won **Qualcomm's Build at the Edge award** at Hack the Six with an autonomous fruit-picking robot pairing a Linux AI chip and a single microcontroller to run LiDAR SLAM navigation, computer vision, and motor control.
- Fine-tuned a YOLOv8 ripeness detector on procedurally generated data and **shrank it 4x** to INT8 so it runs inside the robot's **5-watt** power budget, streaming detections and 3D telemetry live to a React and Three.js dashboard.

4D Voxel Engine | *C++20, OpenGL, GLSL, ASan/TSan*

GitHub

- Engineered a 4D block world explored one 3D slice at a time, meshing **71 million voxels** across nine threads at 6.8 ms/frame by skipping hidden faces between solid blocks, **cutting drawn geometry up to 70x** in dense terrain.
- Optimized greedy meshing into a 4D mesher merging adjacent identical cubes into a single volume, reducing triangle count **39%** and mesh memory **11.6x** while matching an unoptimized implementation on 10,000+ random worlds.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, Java, C#, SQL, Bash, HTML, CSS

Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Three.js

Cloud/Data: AWS, PyTorch, PostgreSQL, MongoDB, Redis, Kafka, Supabase, MySQL, OpenCV, pandas, NumPy

DevOps/Tools: Kubernetes, Docker, Helm, Jenkins, GitHub Actions, pytest, Git, Linux, CI/CD, Jest, Agile, Jira